

AURA V5 — Data Fusion Roadmap & Engineering Journey

1. Objective

The goal of AURA V5 was to transform **heterogeneous real estate datasets** into a **unified, production-grade intelligence system** capable of supporting:

- Property valuation (price_per_m²)
- Investment decision-making
- Market analysis across space and time

The challenge was not just merging data, but building a **consistent, explainable, and scalable fusion pipeline**.

2. Initial Challenge

The datasets were **highly heterogeneous** across multiple dimensions:

Dimension	Problem
Spatial	Grid (Cat5/8) vs Municipality (Cat7)
Temporal	Monthly (Cat3), Quarterly (Cat2), Semi-annual (Cat7)
Structure	Property-level (Cat1) vs Aggregated signals
Keys	No direct join between layers

👉 The core problem became:

How to align ALL datasets across space + time without creating errors or data leakage

3. Fusion Strategy Design

Instead of a simple merge, a **layered architecture** was designed:

```
BASE → ALIGN → TARGET → CONTEXT → BEHAVIOR → FINAL
```

This ensured:

- Controlled joins
- No row explosion
- Clear separation of concerns

◆ 4. Step-by-Step Fusion Pipeline

STEP 1 — Base Layer (Spatial Foundation)

Datasets:

- Cat5 (urban grid intelligence)
- Cat8 (POI / desirability)

Join key:

```
grid_id_250m
```

Result:

- Stable base dataset (~171k grids)
- Pure spatial representation

STEP 2 — Spatial Alignment

Problem:

No direct link between grid and municipality.

Solution:

- Used centroid-based mapping
- Assigned each grid to a municipality

 Output:

```
grid_id → municipality
```

STEP 3 — Target Injection (Ground Truth)

Dataset:

- Cat7 (transaction price per m²)

Join keys:

```
municipality + year + quarter
```

 Critical rule:

- Never join on year alone
- Never join on quarter alone

 Result:

- Creation of a **spatio-temporal panel**
- Each grid expanded across time

STEP 4 — Context Integration

Datasets:

- Cat2 (market dynamics)
- Cat3 (macro)
- Cat4 (socioeconomic)
- Cat1_2 (micro-market signals)

Key alignment:

year + quarter

 Special handling:

- Cat3 aggregated from monthly → quarterly
- Cat1_2 reconstructed using month → quarter logic

 Result:

- Fully contextualized dataset
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STEP 5 — Behavioral Layer

Dataset:

- Cat6 (listing / demand signals)

Process:

- Cleaned
- Aggregated to (year, quarter)
- Integrated safely

 Result:

- Real-world behavior added to system
-

STEP 6 — Cat1 Integration (Final Layer)

Challenge:

- No location
- No time
- No direct mapping

Solution:

- Global aggregation (mean profile)
- Broadcast across dataset

📌 Important outcome:

- Features were constant → automatically removed during cleaning

👉 This validated the robustness of the pipeline.

STEP 7 — Final Cleaning & Validation

Final dataset passed all checks:

- ✓ No duplicates
- ✓ 0% missing values
- ✓ No infinite values
- ✓ Stable row count
- ✓ Valid primary key

📌 Final structure:

```
grid_id × year × quarter → 1.25M rows
```

🧠 5. Key Engineering Principles

1. No Join Explosion

All merges were controlled and validated:

```
Rows before == Rows after
```

2. Strict Key Integrity

All joins enforced:

```
(geo + time) alignment
```

3. No Data Leakage

- Target never used in feature engineering
 - Temporal consistency preserved
-

4. Modular Design

Each dataset handled independently → then fused

👉 Enables future upgrades (e.g., replacing Cat1 easily)

6. Final System Outcome

The result is not just a dataset, but:

```
AURA V5 = Multi-layer real estate intelligence engine
```

It integrates:

- Spatial structure
 - Market behavior
 - Economic context
 - Temporal evolution
-

7. Limitations Identified

- Partial municipality coverage (~83%)
- Cat1 lacks spatial alignment
- No transaction-level granularity (aggregated data)

👉 These are **known and controlled limitations**

8. Conclusion

The AURA V5 fusion process successfully transformed fragmented datasets into a:

- ✓ Clean
- ✓ Scalable
- ✓ Explainable
- ✓ Production-ready system

The pipeline ensures that future improvements (better datasets, finer granularity) can be integrated without redesigning the architecture.

9. Next Step

With the data layer complete, the system is now ready for:

Modeling & AI layer (prediction + decision intelligence)
